

AUTISTUDY — RESEARCH PAPER WRITING GUIDE

Final Edition — Updated for Q1 & Q2 Journal Standards

All Critical Action Items Applied • Correct Frontend & Backend Specs Integrated

Aliza · Tehreem · Zoha · Tayyaba | Intro Due: 25 May | Full Draft: 1 June | Submit: 8 June 2026

WHAT ARE Q1 & Q2 JOURNALS? — READ THIS FIRST

Journals are ranked Q1–Q4 by Scimago (SJR) and Clarivate (JCR) based on Impact Factor and citation influence. Q1 = top 25%, Q2 = top 50%. Both are highly respected. Q1 demands larger samples, stricter stats, and stronger novelty. Q2 is more achievable for a final-year project with a pilot study.

Quartile	Impact Factor	Reviewer Expectations	Realistic for AntiStudy?
Q1	IF ≥ 4.0	Large sample ($N \geq 27$), RCT or strong quasi-experiment, 50+ refs, extensive novelty proof	Possible with full user study
Q2	IF 2.0–4.0	Pilot studies accepted, $N \geq 15$ OK if justified, 40+ refs, clear contribution	✓ Achievable by June 8
Q3	IF 1.0–2.0	Smaller studies accepted, conference-level rigor	Below your target
Q4	IF < 1.0	Minimal bar	Not your target

✓ **CORRECTION APPLIED:** The guide now covers BOTH Q1 and Q2 standards throughout. Q1 requirements are marked 🏆, minimum Q2 requirements are the default text. If you cannot reach Q1 thresholds by June 8, aim for Q2 and explicitly label the paper a Pilot Evaluation Study.

RECOMMENDED Q1 & Q2 JOURNALS FOR AUTISTUDY

Journal	Publisher	IF	Quartile	Scope Fit	Word Limit
Computers & Education	Elsevier	~12.0	Q1	★★★★★	8,000–12,000
IEEE Trans. Learning Technologies	IEEE	~5.0	Q1	★★★★★	6,000–10,000
Expert Systems with Applications	Elsevier	~8.5	Q1	★★★★★	8,000+
Computers in Human Behavior	Elsevier	~9.0	Q1	★★★★★	7,000–10,000
Education & Info Technologies	Springer	~3.8	Q2	★★★★★	6,000–9,000

J. of Special Education Technology	SAGE	~2.5	Q2	★★★★★	5,000–8,000
Applied Sciences (MDPI)	MDPI	~2.7	Q2	★★★	No limit
IEEE Access	IEEE	~3.9	Q2	★★★★★	No limit

 **TIP:** J. of Special Education Technology is an ideal Q2 target — scope is a perfect match for AutoStudy and pilot evaluation studies are explicitly welcomed.

PAGE & WORD BUDGET — CORRECTED

✓ **CORRECTION APPLIED:** Previous edition stated '20–28 pages'. Corrected: 8,000–12,000 words = 12–15 pages in a single-spaced journal template (NOT 30–40 pages, which applies to double-spaced manuscript format). Most journal templates use single-spacing.

#	Section	Words (Q1)	Words (Q2)	Due Date	Assigned To
1	Title, Abstract & Keywords	300–350	250–300	Write last	All
2	Introduction	700–900	600–800	25 May (TOMORROW)	Aliza
3	Related Work	1,200–1,500	900–1,200	31 May	Tayyaba+Aliza+Tehreem
4	System Design & Architecture	1,200–1,500	1,000–1,200	31 May	Zoha
5	Key Features & Modules	800–1,000	600–800	31 May	Tehreem
6	Implementation	600–800	500–700	31 May	Aliza
7	Evaluation & Results	1,500–2,000	1,000–1,500	31 May	Tayyaba
8	Discussion	800–1,200	700–1,000	1 June	Tayyaba
9	Conclusion & Future Work	400–500	300–400	Write last	All
10	References + Extras	—	—	Ongoing	All
	TOTAL	~8,000–10,000	~6,000–8,000	8 June	—

1. Title, Abstract & Keywords

📄 ~0.5 page | Write last
👤 All Members

1.1 Paper Title

For a pilot study targeting Q2, consider appending the study scope to the title to set reader expectations correctly:

Q1 Target: *"AutiStudy: An AI-Powered Virtual Shadow Teacher for Autistic Students in Pakistan"*

Q2 / Pilot Target: *"AutiStudy: A Pilot Evaluation of an AI-Powered Virtual Shadow Teacher for Autistic Students in Pakistan"*

✓ **CORRECTION APPLIED:** Adding 'Pilot Evaluation' to the Q2 title is not a weakness — it is accurate framing that prevents desk-rejection caused by reviewers expecting a full RCT.

1.2 Structured Abstract (250–350 words)

Most Q1/Q2 journals require a structured abstract with labelled headings. Write after all other sections are done.

Label	What to Write	Word Target
Background	ASD prevalence globally (WHO 1:100, CDC 1:36) and in Pakistan; shadow teacher scarcity; STEM learning gap; absence of bilingual AI tools.	40–50
Objective	AutiStudy aims to act as a virtual shadow teacher for autistic students (Grades 4–7) in Pakistan by providing curriculum-aligned STEM tutoring in English and Urdu.	20–30
Methods	Hybrid RAG pipeline (Dense+BM25+RRF+Cross-Encoder) for Science/CS; RAT pipeline for Mathematics; nine-agent agentic AI (LangGraph); multimodal output (TTS, DALL·E, 11 visual aid types); bilingual EN/Urdu. Pilot evaluation: [N] participants, SUS usability scale, pre/post quiz design, Shapiro-Wilk + paired t-test.	60–80
Results	Retrieval Precision@5=87%, MRR=1.00 (RAT maths), +53% text extraction quality. SUS=[X], quiz improvement=[X%], p=[X], Cohen's d=[X]. (Fill after study.)	50–60
Conclusion	AutiStudy demonstrates that curriculum-grounded agentic AI can provide accessible, affordable STEM support for autistic learners in low-resource settings like Pakistan.	30–40

● **CRITICAL:** Do NOT start with 'In this paper...' — start directly: 'Autism Spectrum Disorder (ASD) affects approximately 1 in 36 children globally...'

1.3 Keywords (6–10 terms)

Suggested: Autism Spectrum Disorder; Virtual Shadow Teacher; Retrieval-Augmented Generation; Retrieval-Augmented Thinking; Agentic AI; Special Education; Bilingual NLP; Pakistan; Adaptive Learning; Pilot Evaluation

- Use IEEE Thesaurus terms for IEEE journals; MeSH terms for medical/ASD journals
- Do NOT repeat words already in the title
- Separate with semicolons (IEEE) or commas (MDPI/Springer)

2. Introduction

📄 ~2–3 pages | 700–900 words

👤 Aliza Ali — Due: SUNDAY 25 MAY 2026

31 DEADLINE: The Introduction is due TOMORROW, Sunday 25 May 2026. Write all five paragraphs below today. The retrieval metrics (Precision@5=87%, MRR=1.00) are real and can be included now. The user study placeholders are marked — fill them after the study.

Paragraph 1 — Global & Local Context

- ASD prevalence: WHO (1 in 100), CDC 2023 (1 in 36) — cite both
- Pakistan: growing awareness but no reliable national statistics — cite a Pakistani ASD study
- Shadow teacher concept: one-on-one aide during class — helpful in theory but inaccessible in practice
- Cost: PKR 30,000–80,000/month — cite a local source or institutional report
- Scope limitation: most shadow teachers focus on communication/life skills, not STEM subjects

🏆 **Q1/Q2 STANDARD:** Q1 reviewers expect 3–4 citations in Paragraph 1. Every claim without a citation is an assertion, not evidence.

Paragraph 2 — Gap in Existing Technology

- General AI tutors (Khanmigo, Socratic, Photomath): designed for neurotypical learners, no sensory adjustability

- ASD-specific apps (MITA, Otsimo, Proloquo2Go): address early communication, not STEM at Grades 4–7

✓ **CORRECTION APPLIED:** Explicitly cite Maqсад and Sabaq Foundation here. State their architectural limitation: video-only delivery with no adaptive AI, no Urdu NLP, and no ASD accommodation. This was missing from the previous guide.

- RTL / bilingual NLP gap: standard Western conversational agents (ChatGPT, Gemini) do not correctly handle Urdu RTL text — token allocation for Urdu is higher than Latin scripts in OpenAI models, causing truncation and layout failures
- No existing system combines: ASD-adapted UI + Urdu RTL + curriculum-aligned RAG + Pakistan context

● **CRITICAL:** This paragraph must contain the phrase: 'No prior work has...' — Q1/Q2 reviewers look for this explicit gap statement.

Paragraph 3 — Proposed Solution

"We present AutoStudy, an AI-powered web-based platform that acts as a virtual shadow teacher by providing curriculum-aligned STEM tutoring via a hybrid RAG pipeline and a Retrieval-Augmented Thinking (RAT) framework, coordinated by a nine-agent agentic AI architecture, with multimodal output (text, 11 in-browser visual aid types, DALL·E images, and OpenAI TTS voice narration) and full bilingual support (English and Urdu with RTL rendering) — designed specifically for autistic students in Pakistan at Grades 4–7."

⚠ **NOTE:** Do not describe RAG pipeline steps, agent names, or technical architecture in the introduction. That is Section 3 (Methodology). The intro only says what the system does for users.

Paragraph 4 — Numbered Contributions

Make each contribution specific and measurable. Vague contributions get rejected.

1. AutoStudy — the first AI-powered STEM tutoring system for autistic students in Pakistan combining Urdu bilingual support, curriculum-aligned RAG, and ASD-adapted UI in one platform.
2. A hybrid RAG pipeline (Dense + BM25 + RRF + Cross-Encoder re-ranking with LLaMA Parse ingestion) achieving Retrieval Precision@5 of 87% and Answer Relevance of 91% — improvements of +45% and +36% respectively over OCR-based RAG baselines.
3. A Retrieval-Augmented Thinking (RAT) pipeline for mathematics achieving MRR of 1.00 for procedural queries versus 0.255 for the RAG baseline, via chain-of-thought sub-query decomposition.
4. A nine-agent agentic AI architecture coordinated via LangGraph — including Supervisor, Content, RAG, Critic, Media, Personalisation, Sensory, Recap, and Progress agents — enabling real-time adaptive personalisation.

5. Eleven in-browser visual aid types (EmojiCounting, MathSteps, FactorTree, FractionBar, NumberLine, BarChart, PercentageBar, TimesTable, Geometry, Ratio, and DALL-E concept images) with webcam-based emotion detection for modality adaptation.
6. A pilot usability and learning outcome evaluation with [N] participants, yielding SUS=[X] and quiz improvement of [X%] ($p < 0.05$, Cohen's $d = [X]$). (Fill after study.)

Paragraph 5 — Paper Roadmap

"The remainder of this paper is organised as follows. Section II reviews related work. Section III presents the system architecture and methodology. Section IV describes key features and modules. Section V details implementation. Section VI reports evaluation results. Section VII discusses findings and limitations. Section VIII concludes with future directions."

3. Related Work

📄 ~3–4 pages | 1,200–1,500 words

👤 Tayyaba + Aliza + Tehreem
— Due: 31 May

⚠️ NOTE: Do not write a list of paper summaries. Every subsection must compare papers to each other AND identify a specific gap that AutoStudy fills. End each subsection with a gap sentence.

3.1 AI & Technology for Autism Education

- Cover 6–8 papers: MITA, Otsimo, AuThink, Kaspar robot, Hajjaj et al. ML framework, Nelaka et al. emotion recognition, F.D et al. STEM outcomes
- Compare across: STEM coverage, grade level, user testing on real ASD children, language support
- **Gap sentence:** *"Despite these advances, no reviewed system provides upper-primary STEM learning in a bilingual, low-resource South Asian educational context."*

3.2 LLM-Based Tutoring & RAG Systems

- Original RAG paper (Lewis et al., 2020) — must cite
- Danurata et al. — RAG chatbots using textbooks as knowledge bases (already in your refs [1])
- Salemi & Zamani eRAG — $k=6$ document limit justification (already in refs [4])
- Qiang et al. — knowledge filtering, 88% agreement with human evaluators (already in refs [13])
- Khanmigo, Magic School, Socratic — neurotypical-only, no curriculum grounding
- **Gap sentence:** *"Existing RAG educational systems do not accommodate structured non-English curriculum documents or the specific content needs of neurodiverse learners."*

3.3 Retrieval-Augmented Thinking & Chain-of-Thought

- Wei et al. (2022) — Chain-of-Thought Prompting Elicits Reasoning in LLMs — the seminal CoT paper, MUST cite this
- Luong et al. REFT — reasoning significantly improves multi-step accuracy (referenced in your paper as [?]) — find the correct citation)
- Explain why standard RAG fails for procedural maths: retrieves concept text, cannot verify computation steps
- **Gap sentence:** *"No prior educational RAG system has combined chain-of-thought sub-query decomposition with curriculum vector retrieval for step-by-step mathematics tutoring aligned to a national syllabus."*

3.4 Agentic AI in Education

- Vaccaro et al. [15] multi-agentic LLMs for STEM — already in your refs
- KA-RAG (Gao et al. [21]) — knowledge graph + agentic RAG for educational Q&A
- Williams [22] — human-agentic AI learning architecture
- LangGraph as an orchestration framework for multi-agent educational systems
- **Gap sentence:** *"Multi-agent educational AI has not been applied to ASD-specific tutoring where agents must simultaneously manage sensory UI adaptation, curriculum retrieval, bilingual output, and parental reporting."*

3.5 Multilingual NLP, Urdu & Pakistani EdTech

✓ **CORRECTION APPLIED:** This subsection was previously underdeveloped. The following content directly addresses the action items: (1) cite Maqсад and Sabaq limitations explicitly, and (2) emphasise the RTL/bilingual NLP gap in Western AI systems.

- Maqсад and Sabaq Foundation: well-designed for mainstream students via video-based content, but architecturally limited to passive consumption — no adaptive AI, no NLP, no accommodation for neurodiverse learners
- Taleemabad: curriculum-aligned video platform, no interactive AI or Urdu NLP capabilities
- RTL gap: standard transformer-based language models (GPT, Gemini, LLaMA) tokenize Urdu using subword units calibrated for Latin scripts — Urdu text requires significantly more tokens per word, causing context window overflow and incorrect RTL layout in front-ends not built for bidirectional text
- No existing Pakistani EdTech platform provides adaptive, AI-driven, bilingual support for neurodiverse learners at the upper-primary level
- **Gap sentence:** *"Pakistani EdTech serves general mainstream students through passive video content and does not address the bilingual NLP, RTL rendering, or adaptive learning requirements of autistic students."*

3.6 Comparison Table

Feature	MITA	Otsimo	Khanmigo	ChatGPT	Maqsad	AuThink	KA-RAG	AutiStudy
STEM Gr. 4–7	✗	Partial	✓	✓	✓	✗	✓	✓
ASD-Adapted UI	✓	✓	✗	✗	✗	✓	✗	✓
Urdu / RTL	✗	✗	✗	Partial	Partial	✗	✗	✓
Curriculum RAG	✗	✗	Partial	✗	✗	✗	✓	✓
RAT for Maths	✗	✗	✗	✗	✗	✗	✗	✓
Agentic AI	✗	✗	✗	✗	✗	✗	Partial	✓ (9 agents)
Emotion Detect	✗	✗	✗	✗	✗	✗	✗	✓
Parent Dashboard	✗	Partial	✗	✗	✗	✗	✗	✓
Hallucination-Free	N/A	N/A	Partial	✗	N/A	N/A	✓	✓

● **CRITICAL:** Every row (except AutiStudy) must cite the corresponding paper or product. Uncited comparison tables are flagged by Q1/Q2 reviewers.

4. System Design & Architecture

📄 ~3–4 pages | 1,200–1,500 words

👤 Zoha Khalid — Due: 31 May

4.1 System Overview

- Include Fig. 1: the four-layer architecture diagram (Frontend → FastAPI Backend → AI Core → Agentic Layer)
- Describe each layer in one sentence before expanding in subsections
- Reference all figures: 'As illustrated in Fig. 1...'

🏆 **Q1/Q2 STANDARD:** Figures must be vector (SVG) or ≥300 DPI PNG. Export from draw.io or Figma — never screenshot a diagram.

4.2 Frontend Architecture — CORRECTED FROM TECHNICAL DOCS

✓ **CORRECTION APPLIED:** The frontend stack below is taken directly from the AutiStudy-React technical documentation (autistudy-web v0.1.0). Previous versions of this guide contained inaccuracies (wrong font names, missing libraries). Use only the values below.

Layer	Technology	Version	Justification for Paper
Framework	Next.js (App Router)	14.2.35	SSR enables fast initial load critical for children with short attention spans; file-based routing supports multi-role views (student, parent)
UI Library	React	18	Component-based architecture; supports concurrent rendering for smooth animated UI
Language	TypeScript	5	Compile-time type safety is non-negotiable for a child-facing application where runtime errors disrupt learning sessions
Styling	Tailwind CSS	3.4.1	Utility-first CSS enables rapid accessible UI development; CSS variables enable real-time sensory theme switching without page reload
Animation	Framer Motion	12.38	Declarative animation API allows all transitions to be disabled via a single prop — essential for sensory-averse autistic users
Scroll	Lenis	1.3	Smooth scroll normalisation prevents jarring page jumps that can distress autistic learners
Landing Anim	GSAP	3.15	GPU-accelerated landing page animations; isolated from the tutoring UI so they do not affect sensory settings
Markdown/Math	react-markdown + KaTeX	—	Server-side math rendering prevents layout shift; <code>normalizeMath()</code> handles both $(...)$ and $[...]$ delimiters
Icons	lucide-react	1.8	Consistent, accessible icon set with ARIA label support
Fonts	Quicksand (display) + Inter (body)	Google Fonts	Rounded letterforms in Quicksand are shown to be more readable for learners with dyslexia and ASD
E2E Testing	Playwright	1.59	Automated verification of all critical user flows (login, chat, quiz, parent dashboard)
API Client	Custom typed client (lib/api/client.ts)	—	Typed fetch wrapper with <code>ApiError</code> handling; dual Bearer token model (student: <code>autistudy_token</code> ; parent: <code>autistudy_parent_token</code>)

Key Frontend Routes

Route	Auth	Purpose
/	Public	Landing page with GSAP animations
/login, /signup	Public	Student/parent login; dual role picker; B-Form multipart upload
/dashboard	Student	Stats, subject cards, recent chats, animated greeting
/chat	Student	AI tutor — URL state machine (?subject → ?session); 11 visual aid types
/quiz	Student	Phase-based MCQ flow: pick → chapters → loading → question → result
/analytics	Student	7-day activity chart, subject breakdown, recent attempts
/parent/dashboard	Parent	Child metrics, donut chart, score trend, quiz history
/parent/report	Parent	LLM-generated JSON progress report with section cards

4.3 Backend Architecture — CORRECTED: FastAPI ONLY

✓ **CORRECTION APPLIED:** CORRECTION: Flask has been eliminated. The previous guide incorrectly described a dual FastAPI + Flask backend. The actual AutoStudy backend uses a single FastAPI server (api_server.py) with uvicorn. FastAPI handles both async AI tasks and standard synchronous endpoints natively via a 16-worker ThreadPoolExecutor. Using Flask alongside FastAPI would add unnecessary deployment overhead and dependency conflicts.

Property	Value	Justification
Framework	FastAPI 0.110+	Async-first; auto-generated OpenAPI docs; Python AI ecosystem compatibility; handles both sync and async endpoints natively
ASGI Server	uvicorn	Lightweight; production-ready; supports --reload for development
Default URL	http://127.0.0.1:8000	Sibling repo communication with Next.js on port 3000
CORS Origins	localhost:3000, 3001, 127.0.0.1:3000	Frontend dev and production origins
Thread Pool	16 workers (ThreadPoolExecutor)	Blocking I/O (ChromaDB, EasyOCR, bcrypt) runs via run_in_thread() without blocking the event loop
Auth Model	32-char hex session token (SHA-256)	JWT-like bearer; 7-day TTL; expired tokens deleted on access; bcrypt 12 rounds (SHA-256 legacy migration supported)
Storage	JSON files (data/ directory)	Shared between FastAPI and Streamlit legacy UI; appropriate for research prototype
Image Cache	temp_generated_images/	DALL·E outputs served via static mount

		/api/generated-images/
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REST API Endpoint Groups

Group	Key Endpoints
Public	GET / health, GET /api/chat/config, GET /api/debug/openai-ping (remove before production)
Student Auth	POST /api/auth/register, /login, /logout, GET /api/auth/me, POST /api/auth/child/signup (multipart B-Form)
Parent Auth	POST /api/auth/parent/signup, /login, /logout, GET /api/auth/parent/me
Dashboard	GET /api/users/me/stats, /subjects, /recent-chats, /recent-quizzes
Chat	GET/POST /api/chat/sessions, POST /messages, /image, /speech, /quiz
Quiz & Analytics	GET /api/quiz/chapters, POST /generate, /generate-from-chat, /submit, GET /analytics
Teaching Agent	POST /api/agent/analyze-emotion, /session-summary, /force-modality, GET /api/agent/state, /memory
Media Agent (excluded)	POST /api/agent/run — ReAct GPT-4o function-calling loop (documented separately)

4.4 Data Collection & LLaMA Parse Ingestion

✓ **CORRECTION APPLIED:** Report chunk sizes in BOTH tokens and characters. Urdu text requires significantly more OpenAI tokens per word than Latin text due to subword tokenization — this must be acknowledged in the paper.

- Pakistan SNC curriculum textbooks (PDFs), Grades 4–7: Mathematics, General Science, Computer Science
- OCR attempted first (EasyOCR + OpenCV) → failed: 45% text quality; mathematical equations, Urdu text, and tables were malformed
- LLaMA Parse (LlamaIndex AI-driven document parser) → 98% text quality: preserves LaTeX math notation, headings, tables, and Urdu text structure
- Chunking: semantic chunks with metadata — Chapter Name, Section Title, Page Number, Block Type (Definition / Procedure / Example / Practice / Explanation)
- **Token vs character reporting:** Report each chunk's size in characters (e.g. avg 800 chars) AND estimated OpenAI tokens (e.g. avg 200 tokens for English, avg 350 tokens for equivalent Urdu content). This difference directly explains your k=6 context limit decision.
- Metadata weighting: Procedure and Explanation block types weighted higher for retrieval — report weighting coefficients numerically

🏆 **Q1/Q2 STANDARD:** Add a Dataset Statistics table (Section 6.2): number of books, total pages,

total chunks, avg chunk size in tokens and characters, vocabulary size.

4.5 Hybrid RAG Pipeline (Science & Computer Science)

- Subject-specific pipeline — Science and CS use the same hybrid approach; Maths uses RAT (see 4.6)
- Step 1 — Intent Detection: rule-based keyword classifier identifies query type (definition / example / procedure / problem-solving)
- Step 2 — Parallel Retrieval: Dense semantic search (ChromaDB, all-MiniLM-L6-v2) + BM25 sparse retrieval — weighted 65% dense / 35% BM25 for Computer Science; RRF fusion k=60 for Science
- Step 3 — Reciprocal Rank Fusion (RRF, k=60): merges ranked lists from dense and BM25 retrievers
- Step 4 — Keyword Gating + Metadata Filtering: removes chunks whose block type does not match detected intent
- Step 5 — Cross-Encoder Re-ranking: MS-Marco-MiniLM-L6-V2 re-ranks top 40 candidates → top 6 selected (k=6 justified by Salemi & Zamani [4])
- Step 6 — GPT-4o Mini Generation: top 6 chunks injected as grounding context for curriculum-aligned response
- Include Fig. 2: RAG pipeline diagram at ≥300 DPI

4.6 RAT Pipeline (Mathematics) — NOTE on Production

✓ **CORRECTION APPLIED:** The RAT pipeline achieves MRR=1.00 in evaluation, but is DISABLED in the production API for latency reasons. Acknowledge this explicitly in the paper: report the evaluation metric from offline testing, and note that the production system uses the optimised hybrid RAG for Maths queries with a 65%/35% Dense+BM25 split pending latency optimisation of RAT.

- Step 1 — CoT Decomposition: GPT-4o Mini breaks the problem into sub-questions before any retrieval
- Step 2 — Sub-query Vector Search: each sub-question independently queried against ChromaDB
- Step 3 — Calculation Validation: retrieved content used to verify each intermediate step
- Step 4 — Step-by-step generation: GPT-4o Mini synthesises a textbook-aligned explanation
- Include Fig. 3: RAT pipeline diagram at ≥300 DPI

4.7 Nine-Agent Agentic Architecture

Agent	Role	Technology
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Supervisor	Coordinates all agents via LangGraph StateGraph; routes queries to RAG or RAT pipeline	LangGraph
Content	Intent detection; builds retrieval search plan	Rule-based + GPT-4o Mini
RAG	Hybrid retrieval from ChromaDB + BM25 + cross-encoder re-ranking	ChromaDB, BM25, MS-Marco-MiniLM
Critic	Evaluates answer accuracy, clarity, and age-appropriateness via chain-of-thought reasoning	GPT-4o Mini CoT
Media	Image generation (gpt-image-1.5 → gpt-image-1 → dall-e-3 fallback chain) + TTS (tts-1, alloy/nova)	OpenAI API
Personalisation	Adjusts difficulty based on quiz performance history stored in agent_memory JSON	GPT-4o Mini + JSON
Sensory	Adjusts UI in real time: font size, contrast, brightness, animation toggle, sound effects	React state + CSS variables
Recap	Summarises past conversations for session continuity using GPT-4o Mini	GPT-4o Mini
Progress	Stores activities in JSON for long-term memory; feeds parent dashboard	FastAPI + agent_memory/{md5}.json

4.8 Security — Prompt Injection Defence (NEW SECTION)

✓ **CORRECTION APPLIED:** This section was missing from the previous guide. Q1/Q2 reviewers in AI education papers increasingly expect a security discussion, especially for systems used by children.

- The nine-agent architecture inherently defends against prompt injection by design: the Critic agent evaluates every generated answer for accuracy, clarity, and age-appropriateness before it reaches the child — a malicious injection that bypasses the RAG retrieval would be flagged by the Critic agent as irrelevant or off-curriculum
- Intent detection (Content agent): rule-based keyword classifier filters queries before they reach the LLM — queries classified as out-of-scope (not matching Maths, Science, or Computer Science intents) are rejected with a curriculum-boundary message
- Subject-based keyword gating: chunks retrieved from unrelated subjects are filtered before LLM generation
- System prompt isolation: each agent has a fixed system prompt injected at the API level; user messages are passed as user-role content only, preventing role confusion attacks
- Out-of-textbook detection: RAG relevance scorer (threshold 0.35 with subject keywords; 0.55 without) rejects queries that do not match the curriculum — reported at 85% detection rate vs. 20% for the baseline

- Child safety: no web-browsing tool, no code-execution tool, and no external API access is available to student-facing agents — the attack surface is limited to the curriculum knowledge base

 **Q1/Q2 STANDARD:** For a Q1 journal like Computers & Education, consider adding a dedicated threat model table: list injection types (direct prompt injection, indirect via retrieved content, jailbreak attempts) and how each is mitigated.

5. Key Features & Modules

 ~2–3 pages | 800–1,000 words

 Tehreem Ghulam Nabi —
Due: 31 May

5.1 AI Tutor Chat

- English or Urdu input; pipeline selected automatically (RAG for Science/CS, RAT-mode for Maths evaluation)
- KaTeX renders fractions, algebra, and geometry inline; `normalizeMath()` normalises `\\(...\\)` and `\\[...\\]` delimiters
- Framer Motion 12.38 fade-in for responses — toggleable off for sensory-averse users
- Optimistic UI: user message appears instantly with rollback on API error
- Config banner displays when `tutor_configured === false` (OpenAI key not loaded)

 **Q1/Q2 STANDARD:** For every feature, state the ASD design rationale with a citation. E.g., 'Consistent response animations reduce unexpected UI changes, which has been shown to lower distress in autistic learners [CITE].'

5.2 Eleven In-Browser Visual Aid Types

✓ **CORRECTION APPLIED:** The previous guide described 'SVG animations' vaguely. The correct technical term is '11 in-browser visual aid types', all rendered client-side without a network call after the initial API response.

Visual Aid Type	Trigger Condition	ASD Rationale
EmojiCounting	Small whole-number $+/-/ \times$ (≤ 10)	Concrete visual counting supports pre-abstract learners
MathStepCard	Fractions, algebra, long division	Step-by-step reduces cognitive load
FactorTree	HCF, LCM, prime factors	Visual decomposition of abstract relationships
FractionBar	Fraction + visual keyword	Proportional bars make part-whole relationships concrete

NumberLine	Integers, negatives	Spatial representation of abstract number concepts
BarChart	Chart/graph keywords	Familiar visual for data comparison
PercentageBar	Percent questions	Visual proportion without decimal abstraction
TimesTable	Times table keywords	Structured grid for pattern recognition
Geometry	Shapes, perimeter, area	Spatial visualisation of geometric properties
Ratio	Ratio/proportion	Comparative bars for relational reasoning
DALL·E Image	Default concept questions	AI-generated contextual images for abstract concepts

5.3 Webcam Emotion Detection & Modality Adaptation

- EmotionCamera component captures JPEG frames at 320×240, 4-second intervals via getUserMedia
- Frames sent to POST /api/agent/analyze-emotion → GPT-4o Mini Vision classifies: happy, focused, neutral, confused, frustrated, no_face
- Modality ladder: text → text+image → text+image+voice → step_by_step
- ≥2 consecutive confused/frustrated → escalate one level; ≥3 consecutive happy/focused → de-escalate
- Camera consent: localStorage key autistudy_camera_consent (granted | denied | pending); one-time consent modal with 1.5s delay
- Cross-session memory (agent_memory.py): tracks struggling topics, mastered topics, modality success rates per student

5.4 Adaptive Quiz System

- MCQ generation via GPT-4o Mini (gpt-4o-mini, JSON mode); 4 options, correct answer validated in options list
- Sources: by chapter (quizApi.chapters()) or from recent chat transcript (generate-from-chat)
- Star awards: 90%+ → 5 stars, 80%+ → 4, 70%+ → 3, 60%+ → 2, else 1
- Grade-subject mapping: Grades 4–5: Maths + General Science; Grades 6–7: + Computer Science

5.5 Parent-Teacher Dashboard & AI Report

- Separate parent role with isolated session tokens (autistudy_parent_token); parent sees child metrics only
- Dashboard: favourite subject, score trend, speed analysis (avg sec/question), consistency score, improvement metric (recent vs older half)
- AI Report: GPT-4o Mini JSON mode (~1200 tokens) — sections: overview, strengths, improve, tips, next goals; overall rating: Excellent/Good/Developing/Needs Support

- B-Form OCR onboarding: EasyOCR (English CPU) + OpenCV preprocessing (grayscale, upscale 1500px, adaptive threshold, dilate) extracts parent CNICs; validates NADRA B-Form format

5.6 Accessibility & Sensory Settings

- Font size: 4 levels. Contrast mode: high-contrast dark theme. Focus mode: hides non-essential UI. Animation toggle: disables Framer Motion globally.
- Language toggle: English ↔ Urdu with RTL layout flip via LocaleProvider + React Context
- Settings persisted to localStorage via SettingsContext; applied on mount without flash of unstyled content

 **Q1/Q2 STANDARD:** State which WCAG 2.1 success criteria each feature addresses: font size → SC 1.4.4 (Resize Text); contrast → SC 1.4.3 (Contrast Minimum); animation → SC 2.3.3 (Animation from Interactions).

6. Implementation

 ~2 pages | 600–800 words
 Aliza Ali — Due: 31 May

6.1 Full Technology Stack

Present the complete stack as a table. Use the exact values from the frontend and backend technical documents.

Category	Technology	Purpose
Frontend Framework	Next.js 14.2.35 (App Router)	SSR, routing, project structure
UI Library	React 18	Component-based UI
Language	TypeScript 5	Type-safe frontend code
Styling	Tailwind CSS 3.4.1	Utility-first responsive styling
Animation	Framer Motion 12.38 + GSAP 3.15 + Lenis 1.3	UI transitions, landing animations, smooth scroll
Math Rendering	react-markdown + KaTeX + remark-math	Curriculum math symbols and equations
Icons	lucide-react 1.8	UI icons with ARIA support
Fonts	Quicksand + Inter (Google Fonts)	Accessible, readable typefaces
i18n	Custom LocaleProvider (React Context)	English / Urdu (RTL) switching
E2E Testing	Playwright 1.59	Automated critical-flow verification

Backend	FastAPI 0.110+ (uvicorn, 16-worker ThreadPoolExecutor)	REST API server — single framework
LLM	OpenAI GPT-4o Mini	AI tutor, quiz generation, reports, emotion analysis
Image Generation	gpt-image-1.5 → gpt-image-1 → dall-e-3 (fallback chain)	Visual aids for complex concepts
TTS	OpenAI TTS-1 (alloy for EN, nova for UR)	Voice narration
Embeddings	sentence-transformers/all-MiniLM-L6-v2	Semantic chunk embeddings
Reranker	cross-encoder/ms-marco-MiniLM-L6-v2	Cross-encoder re-ranking of top 40 candidates
Vector Store	ChromaDB (collection: ptb_textbooks)	Embedding storage and similarity retrieval
Sparse Retrieval	rank_bm25 + numpy	Keyword-based exact matching
Agent Orchestration	LangGraph	Nine-agent state machine coordination
OCR	EasyOCR + OpenCV + NumPy	B-Form CNIC extraction and parent verification
Security	bcrypt (12 rounds)	Password hashing; SHA-256 legacy migration
Storage	JSON files (data/ directory)	Lightweight file-based persistence — shared by FastAPI and Streamlit legacy
Auth	32-char hex session tokens (7-day TTL)	Child / Parent role-based login

6.2 Dataset Statistics (Q1/Q2 Mandatory)

Fill in real values from your ChromaDB collection and book corpus before submission.

Statistic	Value (Fill In)
Subjects covered	Mathematics, General Science, Computer Science
Grade levels	4, 5, 6, 7
Number of textbooks ingested	[X] books
Total pages	[X] pages
Total chunks created	[X] chunks
Avg chunk size (English)	~[X] chars / ~[X] tokens
Avg chunk size (Urdu-mixed)	~[X] chars / ~[X] tokens (higher token count due to subword tokenization)

BM25 vocabulary size	[X] unique terms
ChromaDB collection size	[X] embeddings
Block type distribution	Definition [X]%, Procedure [X]%, Example [X]%, Practice [X]%, Explanation [X]%
Grade 7 Maths status	math_parse_7.md exists on disk but is NOT in BOOK_MAP — acknowledged as a known limitation

● **CRITICAL:** The Grade 7 Maths book (math_parse_7.md) exists but is not registered in BOOK_MAP. This means Grade 7 Maths quiz chapters are unavailable. State this explicitly in the implementation section as a known limitation — do not hide it.

6.3 Engineering Decisions & Justifications

- FastAPI only (no Flask): FastAPI handles both async AI tasks and synchronous utility endpoints natively via ThreadPoolExecutor — a dual framework adds deployment overhead and dependency conflicts without benefit
- k=6 context limit: justified by Salemi & Zamani [4] — too many retrieved documents degrade LLM generation quality
- LLaMA Parse over OCR: +53% text extraction quality, +65% math symbol accuracy — validated by your own ablation (Table II)
- GPT-4o Mini over GPT-4o: ~10x cost reduction with acceptable quality for K-7 curriculum content — validated by 91% answer relevance score
- JSON over SQL: appropriate for a research prototype; avoids infrastructure overhead; both FastAPI and Streamlit share the same files
- gpt-image-1.5 → gpt-image-1 → dall-e-3 fallback chain: ensures image generation availability across API availability changes

✓ **CORRECTION APPLIED:** The /api/debug/openai-ping endpoint must be removed before production deployment. Note this explicitly in the paper.

6.4 Reproducibility Statement

Prepare one of the following before submission:

- Option A: 'The source code and anonymised evaluation dataset are available at: [GitHub/Zenodo URL]'
- Option B: 'The source code is available upon request from the corresponding author pending institutional data governance approval.'
- Option C (if sharing is restricted): 'Due to privacy constraints involving minor participants, the raw dataset is not publicly available. The system prompt templates, evaluation scripts, and retrieval benchmark queries are available at [URL].'

7. Evaluation & Results

📄 ~4–5 pages | 1,500–2,000 words
👤 Tayyaba Sikandar — Due: 31 May

★ **MOST CRITICAL SECTION. Weak evaluation = rejection at any quartile.**

7.1 Ethics & IRB Approval

● **CRITICAL:** IRB approval is required before conducting ANY study involving children. Apply to UMT's Institutional Review Board immediately. Include the approval number in the paper: 'This study was approved by the IRB of the University of Management and Technology (Approval No. [XXX]).' Without this, the paper will be desk-rejected by all Q1 and Q2 journals.

- Obtain written informed consent from parents/guardians
- Obtain verbal assent from children where developmentally appropriate
- State data handling: chat logs stored locally, no third-party sharing except OpenAI API calls, CNIC data not retained after registration

7.2 Pilot Evaluation Study Framing

✓ **CORRECTION APPLIED:** CORRECTION: The previous guide described a full evaluation with a control group. Since adding a control group is not feasible before June 8, explicitly frame the evaluation as a Pilot Evaluation Study throughout the paper. This is a standard, accepted framing in Q2 journals and some Q1 journals for initial system validation.

Add this sentence to the start of Section 7:

"Given the time and access constraints of recruiting autistic participants in Pakistan, this study is framed as a pilot evaluation to establish initial feasibility and usability evidence, following the conventions of pilot studies in educational technology research [CITE a pilot study methods paper]."

- Label all results as 'preliminary' or 'pilot' — do not overclaim generalisability
- Acknowledge the absence of a control group explicitly in Section 8.4 (Limitations)

7.3 Evaluation Framework

Dimension	What Is Measured	Method	Min N (Q2)
1. Retrieval Quality	Technical pipeline accuracy	Precision@5, Recall@5, MRR, nDCG@5 (done)	N/A — benchmark

			queries
2. Learning Outcomes	Do children learn better with AutoStudy?	Pre-test / Post-test + Shapiro-Wilk + paired t-test	N≥15
3. Usability	Is the system usable for ASD learners?	SUS (parents/teachers) + child-adapted scale	N≥10 per group
4. Qualitative	What do parents think in their own words?	Thematic analysis of open-ended feedback	N≥8 parent responses

7.4 Learning Outcome Assessment — Shapiro-Wilk + t-test

✓ **CORRECTION APPLIED:** CORRECTION: The Shapiro-Wilk normality test must be run BEFORE applying the paired t-test. This was missing from the previous guide. If the data is non-normal ($p < 0.05$ on Shapiro-Wilk), use the Wilcoxon signed-rank test instead.

- Pre-test: 10-question curriculum quiz on target topic (e.g., Grade 5 fractions) before using AutoStudy
- Intervention: 2–3 sessions of using AutoStudy on the same topic
- Post-test: equivalent 10-question quiz on the same topic

Step	Statistical Test	Tool	Decision Rule
1. Check normality	Shapiro-Wilk test on pre and post scores	<code>scipy.stats.shapiro()</code>	If $p > 0.05 \rightarrow$ data is normal $\rightarrow$ use paired t-test. If $p < 0.05 \rightarrow$ use Wilcoxon signed-rank.
2a. Normal data	Paired t-test	<code>scipy.stats.ttest_rel()</code>	Report t-statistic, degrees of freedom, p-value. Need $p < 0.05$.
2b. Non-normal data	Wilcoxon signed-rank test	<code>scipy.stats.wilcoxon()</code>	Report W-statistic and p-value. Need $p < 0.05$.
3. Effect size	Cohen's d	$(M_{\text{post}} - M_{\text{pre}}) / SD_{\text{pooled}}$	Small=0.2, Medium=0.5, Large=0.8. Report alongside p-value.
4. Report	Improvement %	$(\text{post} - \text{pre}) / \text{pre} \times 100$	Practical significance for reader interpretation.

7.5 Usability Evaluation — SUS + Qualitative Thematic Analysis

✓ **CORRECTION APPLIED:** CORRECTION: Qualitative thematic analysis of open-ended parent feedback must be added to complement SUS scores. This was missing from the previous guide.

Q1/Q2 reviewers call this 'triangulation' — using multiple methods to validate the same finding.

SUS (Quantitative)

- Administer the 10-item SUS questionnaire to parents and teachers after using AutoStudy
- For young autistic children: use a 5-item child-adapted scale with smiley-face response options
- Report: mean SUS score, standard deviation, and interpretation (≥ 68 = above average; ≥ 80 = excellent)
- Include a bar chart (Fig. 10) showing scores by participant group
- Cite: Brooke, J. (1996). 'SUS: A quick and dirty usability scale'

Thematic Analysis of Parent Open-Ended Feedback (Qualitative)

- Add 2–3 open-ended questions to the SUS session: 'What did your child enjoy most?', 'What was difficult?', 'Would you use this at home?'
- Conduct inductive thematic analysis (Braun & Clarke, 2006 — cite this): read responses, generate codes, group into themes
- Report 3–5 themes with supporting quotes (anonymised). Example themes: 'Independence in learning', 'Engagement through visuals', 'Language accessibility for Urdu-speaking families'
- Include a small thematic map or table showing codes → themes

 **Q1/Q2 STANDARD:** Triangulation (SUS + thematic analysis) significantly strengthens Q1/Q2 papers on educational systems. It shows the system works quantitatively AND that users genuinely value it.

7.6 Sample Size & G*Power Justification

✓ **CORRECTION APPLIED:** CORRECTION: The previous guide left sample size calculation open. The specific parameters are now stated below. Run G*Power before your study begins.

G*Power Parameters for Paired t-test:

- Effect size: Cohen's $d \geq 0.6$ (medium-to-large — justified by similar ASD+AI intervention studies showing $d=0.5-0.8$)
- Statistical power: $1-\beta = 0.80$ (standard minimum for educational research)
- Significance level: $\alpha = 0.05$
- Test type: Two-tailed paired t-test

Result: Minimum $N \approx 24$ participants (run G*Power to confirm exact value)

- If $N < 24$: label the paper explicitly as a Pilot Study and cite 3 similar pilot papers with

comparable N to justify

7.7 Retrieval Quality Results (Expand Existing)

- You already have Precision@5, Recall@5, MRR, nDCG@5 across 3 subjects × 4 grade levels — this is the strongest part of your current paper
- ADD: number of test queries used per subject per grade (Q2 reviewers check this)
- ADD: statistical significance test comparing Enhanced RAG vs. Traditional RAG (Wilcoxon or paired t-test on the metric scores — not just percentages)
- ADD: inter-rater agreement (Cohen's Kappa) if any relevance judgements were made by human raters

 **Q1/Q2 STANDARD:** Your MRR jump from 0.255 → 1.00 for RAT maths is your strongest single result. Dedicate a full paragraph to explaining WHY: CoT decomposition aligns perfectly with the Procedure block type in your metadata — retrieved chunks are always procedural steps, not concept definitions, so the correct answer is always ranked first.

7.8 Required Figures & Tables

Item	Description	Status
Fig. 1	Full system architecture (4 layers)	☐ Export at ≥300 DPI
Fig. 2	RAG pipeline diagram	☐ Re-export from existing at ≥300 DPI
Fig. 3	RAT pipeline diagram	☐ Re-export from existing at ≥300 DPI
Fig. 4	Nine-agent architecture	☐ Re-export from existing at ≥300 DPI
Fig. 5–8	Precision, Recall, MRR, nDCG@5 bar charts	☐ Re-export at ≥300 DPI
Fig. 9	Pre-test vs. Post-test scores	☐ Add after user study
Fig. 10	SUS scores by participant group	☐ Add after user study
Fig. 11	Thematic map of parent feedback	☐ Add after thematic analysis
Table I	Related systems comparison	✓ Expand with new columns
Table II	System-level performance (existing)	✓ Keep — good as is
Table III	Dataset statistics	☐ Add in Section 6.2
Table IV	Participant demographics	☐ Add after user study
Table V	Statistical test results (t/Wilcoxon, p, d)	☐ Add after user study

8. Discussion

~2–3 pages | 800–1,200 words
Tayyaba Sikandar — Due: 1 June

8.1 Interpretation of Retrieval Results

- OCR → LLaMA Parse: +53% text quality shows document parsing is the critical bottleneck in curriculum RAG — connect to Salemi & Zamani [4]
- MRR = 1.00 for RAT: explain the mechanism — CoT decomposition generates sub-queries that match Procedure block types in metadata, so procedural chunks are always ranked first
- Precision@5 Maths RAT (0.520) vs. RAG baseline (0.153): discuss why dense-only retrieval fails for structured mathematical content

 **Q1/Q2 STANDARD:** For every finding: Result → Interpretation → Connection to prior literature → Implication for the field. A paragraph with only a number is insufficient for Q1/Q2.

8.2 Interpretation of Learning & Usability Results (Add After Study)

- What does [X%] quiz improvement mean in the context of ASD learners who need repeated exposure?
- How does your SUS score compare to similar EdTech systems in the literature?
- What themes emerged from parent feedback? What design decisions do they validate?

8.3 Strengths

- First AI system combining curriculum-aligned RAG + Urdu RTL + ASD-adapted UI in Pakistan
- RAT pipeline solves the known limitation of RAG for mathematical procedural queries
- Nine-agent architecture enables a level of personalisation not available in existing ASD tutoring tools
- Prompt injection defence through intent classification and Critic agent evaluation
- Qualitative + quantitative triangulation provides converging evidence of usability

8.4 Limitations (Write Honestly — Reviewers Respect This)

✓ **CORRECTION APPLIED:** Frame the absence of a control group as a limitation here, and explicitly state the 'Pilot Study' scope.

- Pilot scope: this is an initial feasibility evaluation — results should be interpreted as preliminary evidence pending a larger randomised controlled trial
- No control group: without a randomised control group, quiz improvement cannot be attributed solely to AutoStudy (practice effect, parental assistance are potential confounds)
- JSON storage: not scalable beyond ~1,000 users — production deployment requires a relational database
- RAT disabled in production: the RAT pipeline (MRR=1.00) is currently disabled in the live API due to latency — users experience the hybrid RAG for maths pending latency optimisation
- Grade 7 Maths: math_parse_7.md exists on disk but is not registered in BOOK_MAP — Grade 7 Maths quiz chapters are unavailable in the current build
- API dependency: OpenAI API costs and internet connectivity requirements limit deployment in low-infrastructure areas
- No speech-to-text input: children must type — barrier for non-verbal or low-literacy learners
- ASD heterogeneity: not tested across ASD severity levels (Level 1/2/3) — findings may not generalise across the full spectrum

8.5 Ethical Considerations

- Child data privacy: chat logs stored locally; no third-party sharing except OpenAI API calls (governed by OpenAI's data processing agreement)
- CNIC OCR: parent CNIC extracted from B-Form is used for registration verification only — not retained in plaintext after account creation
- AI bias: GPT-4o Mini carries biases from its training corpus; curriculum-grounding via RAG mitigates out-of-curriculum responses but does not eliminate linguistic or cultural bias
- Dependence risk: AutoStudy is designed to supplement, not replace, human shadow teachers and special education professionals

9. Conclusion & Future Work

~1 page | 400–500 words

All Members — Write Last

9.1 Conclusion Structure

7. Restate the problem: shadow teacher scarcity + STEM gap for ASD children in Pakistan.
8. What AntiStudy provides: curriculum-aligned STEM tutoring via RAG+RAT, nine-agent personalisation, multimodal output, Urdu RTL support.
9. Technical result: Retrieval Precision@5=87%, MRR=1.00 (RAT maths), +65% math symbol accuracy vs. baseline.
10. Pilot user study result: SUS=[X], quiz improvement=[X%] ($p < 0.05$, Cohen's d =[X]) — fill after study.
11. Broader claim: this pilot demonstrates that AI can provide accessible, affordable STEM support for neurodiverse learners in low-resource settings — warranting a full-scale longitudinal evaluation.

⚠ NOTE: Do NOT introduce new information or new citations in the conclusion. Every sentence must refer back to something already stated in the paper.

9.2 Future Work

- Full randomised controlled trial ($N \geq 50$, with control group) to establish causal learning outcomes
- Register Grade 7 Maths (math_parse_7.md) in BOOK_MAP and expand coverage to Grades 8–10
- Enable RAT pipeline in production by optimising latency (caching, async sub-query parallelisation)
- Integrate speech-to-text input (OpenAI Whisper) to remove the typing barrier for non-verbal learners
- Integrate facial emotion recognition (OpenCV + Swin Transformer [19]) into real-time modality decisions
- Mobile app (iOS/Android) with offline support using quantized local LLMs (e.g., Mistral 7B via llama.cpp)
- Additional regional languages: Punjabi, Pashto, Sindhi
- Replace JSON storage with PostgreSQL for production-scale deployment
- Remove /api/debug/openai-ping endpoint before any production deployment
- Longitudinal study (full school year) tracking learning progress and shadow teacher dependency reduction

10. References

~2–3 pages | Target:
50–80 refs
All Members — Ongoing

● **CRITICAL:** Your current paper has ~24 references. Q1 requires 50–80, Q2 requires 40–55. You need to find 20–55 more papers immediately.

Reference Gap Table — What You Must Find

Topic	Search Terms	Papers Needed	Priority
ASD + Pakistan prevalence	'autism prevalence Pakistan', 'ASD South Asia'	3–4	● HIGH
Shadow teacher / paraprofessional	'shadow teacher autism', 'paraprofessional ASD'	3–4	● HIGH
CoT / chain-of-thought reasoning	Wei et al. 2022 CoT paper — MUST cite	2–3	● HIGH
Shapiro-Wilk + normality testing	'normality test educational research'	1–2	● HIGH
Thematic analysis methods	Braun & Clarke 2006 — MUST cite for qualitative	1	● HIGH
Pilot study methodology	'pilot study educational technology'	2–3	● HIGH
Maqsad / Sabaq / Taleemabad	Company websites + any published evaluations	2–3	MEDIUM
Urdu NLP / RTL text	'Urdu NLP', 'RTL interface', 'Urdu tokenization'	3–4	MEDIUM
Hybrid RAG / BM25 + Dense	'hybrid retrieval RAG', 'BM25 dense fusion'	3–4	MEDIUM
SUS methodology	Brooke 1996 + 2 validation studies	2–3	MEDIUM
WCAG accessibility autism	'WCAG autism', 'web accessibility neurodivergent'	2–3	MEDIUM
LangGraph / agentic AI	'LangGraph education', 'multi-agent LLM'	2–3	MEDIUM
LLM hallucination	'hallucination LLM', 'grounding language models'	2–3	LOW
BERT / Transformer fundamentals	Vaswani et al. 2017, Devlin et al. 2019	2	LOW

Mandatory Pre-Reference Sections

Section	What to Write
Data Availability Statement	'The source code is available at [URL]. The evaluation dataset contains data from minor participants and is available upon reasonable request subject to institutional ethics approval.'
Author Contributions (CRediT)	Aliza Ali: Conceptualization, Writing – Original Draft, Software (Frontend). Tehreem Ghulam Nabi: Writing – Original Draft, Features. Zoha Khalid: Software (System Architecture), Visualization. Tayyaba Sikandar: Investigation, Formal Analysis, Writing – Review & Editing.
Acknowledgements	'The authors thank Dr Shaista Habib and Dr Rimsha Anam for their supervision. The authors also thank the special education teachers and autistic students who contributed to the evaluation. This research was conducted at the Department of Artificial Intelligence, University of Management and Technology, Lahore, Pakistan.'

AutiStudy addresses a genuine, under-served problem. A Q2 publication is well within reach by June 8 if the team starts writing today.